

Experimental Assessment of Non-Contact Sensing Technologies for Remote River Level Monitoring

Student: Joud Aloqla · Mentor: Prof.Ermanno Pietrosemoli

KAUST Academy, KAUST · The Abdus Salam International Centre for Theoretical Physics (ICTP)

Introduction

River water level data is critical for flood warning and water resource management. But the rivers that need monitoring most are often in remote areas, without cellular coverage, and require a low-cost system that can run independently on limited battery power.

This project builds a complete monitoring system using LoRaWAN for long-range, low-power data transmission, comparing three non-contact sensors (Ultrasonic, Radar, and LiDAR) for accuracy, performance, limitations, and overall reliability in remote river settings.

Objectives

- Evaluate and compare the performance of Ultrasonic, Radar, and LiDAR sensors under various field conditions.
- Implement a robust communication architecture to transmit real-time data wirelessly from remote areas.
- Optimize power consumption and extend battery life using a dynamic data transmission strategy.

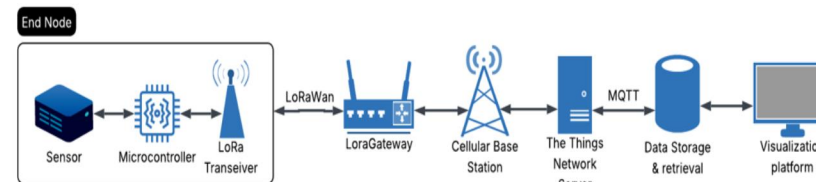
Adaptive Transmission Strategy

The system dynamically adjusts the transmission interval based on river activity, maximizing the sleep interval when the water level is stable and increasing frequency during rapid changes, reducing power consumption and cost without missing critical flood data.

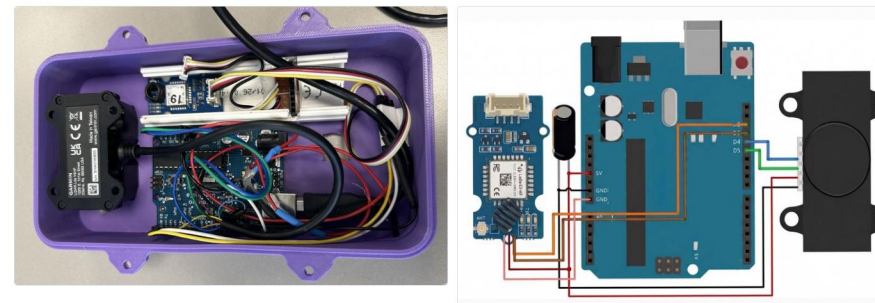
Condition	Change / reading	Interval
Stable flow	<1%	15 min
Moderate activity	1–3.5%	5 min
Rapid change	>3.5%	1 min

Methodology

System Architecture



Sensor	Wave / Freq.	Range	Beam
Ultrasonic	Acoustic, 40 kHz	0.25–10 m	~60°
Radar	EM, 60 GHz	0.3–12 m	8°
LiDAR	EM, 905 nm	up to 40 m	~0.5°



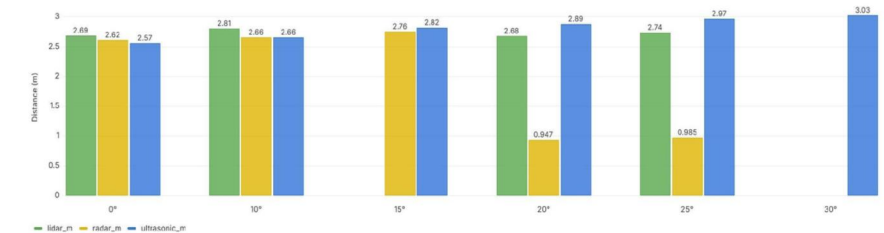
The custom LiDAR node, opened to show its wiring (left), and the wiring diagram (right).

Timavo River Experiment

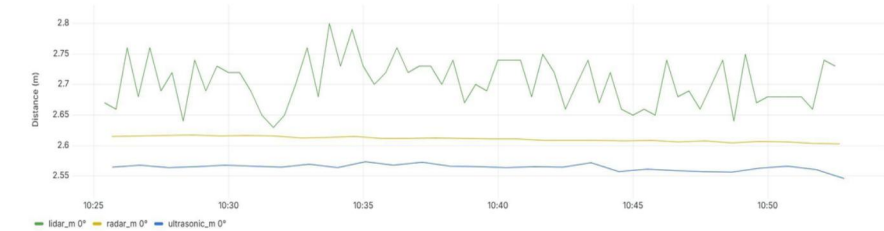


Results

Average Distance Measured by Each Sensor at Different Tilt Angles



Sensors Performance and Measurement Variations Over Time



- Radar:**
Strength: highly stable and accurate under normal conditions. **Limitation:** fails sharply (–66.5% error) when tilted beyond its 15° rated limit.
- Ultrasonic:**
Strength: maintains high accuracy even when tilted up to 30°. **Limitation:** the wide 60° beam needs a clear installation space, nearby structures cause false readings.
- LiDAR:**
Strength: longest range, with accuracy improving in turbid water. **Limitation:** penetrates calm, clear water instead of reflecting; its narrow spot-beam captures instant surface noise, causing variation.

Future Work

- Sensors were not tested under turbulent flow. *Next step:* repeat the trial during higher-flow or post-rainfall conditions, when accurate readings matter most for flood warning.
- Battery and solar performance under the adaptive scheme were not evaluated. *Next step:* long-term field testing to assess power sustainability and reliability.

References

- J. D. Paul, W. Buytaert, N. Sah, “A Technical Evaluation of Lidar-Based Measurement of River Water Levels,” *Water Resources Research* 56.4 (2020), e2019WR026810.
- J. N. Kabi et al., “Low cost, LoRa based river water level data acquisition system,” *HardwareX* 14 (2023), e00414.